

SAO LUIS, Brazil

WMO: 822810

Lat: 2.585S

Lon: 44.234W

Elev: 178

StdP: 14.60

Time Zone: -3.00 (W03)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
4	73.1	73.6	64.0	89.9	88.1	65.2	93.8	87.0	18.6	84.1	16.7	84.8	4.8	60	0.424

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
11	11.5	91.9	77.9	91.4	77.7	90.0	77.0	81.7	86.8	80.9	86.4	80.2	85.9	13.3	60

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
80.6	159.9	84.7	79.2	152.7	83.4	78.9	151.2	83.1	45.7	86.6	44.8	86.7	44.0	86.9	88.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 21.0	19.0	17.6	70.1	94.8	4.1	2.5	67.2	96.6	64.8	98.1	62.5	99.5	59.6	101.3		
(5)			67.8	82.6	3.3	3.1	65.4	84.8	63.5	86.6	61.6	88.3	59.2	90.5		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	82.7	82.3	81.5	81.3	81.3	82.1	82.2	82.2	83.1	83.7	84.2	84.4	84.0		
	DBStd	2.06	2.23	2.38	2.12	1.92	1.81	1.47	1.54	1.33	1.26	1.15	1.24	1.95		
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0		
	HDD65	0	0	0	0	0	0	0	0	0	0	0	0	0		
	CDD50	11931	1000	882	969	938	994	966	998	1025	1012	1061	1031	1055		
	CDD65	6456	535	462	504	488	529	516	533	560	562	596	581	590		
	CDH74	65663	5463	4411	4638	4354	5017	5037	5165	5721	6008	6594	6577	6678		
(13)	CDH80	21759	1743	1293	1255	1167	1524	1613	1661	1954	2129	2419	2480	2521		
(14)	Wind	WSAvg	8.7	9.0	7.6	6.7	5.6	5.9	6.3	7.1	9.3	11.5	12.3	12.2		
(15) Precipitation	PrecAvg	70.4	7.4	10.1	12.8	14.1	10.5	5.8	4.1	1.2	0.3	0.2	0.4	2.4		
	PrecMax	105.4	17.7	19.0	22.7	23.7	21.2	10.3	9.6	3.2	1.4	2.1	1.6	10.4		
	PrecMin	39.2	0.7	3.7	4.8	6.1	3.1	0.9	1.0	0.3	0.0	0.0	0.0	0.8		
	PrecStd	17.1	4.9	4.7	4.3	5.4	4.4	2.4	2.0	0.8	0.3	0.4	0.4	2.0		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	91.7	91.5	90.1	90.4	91.5	91.3	91.6	91.8	93.2	93.3	93.3	93.4		
		MCWB	78.5	78.5	78.9	78.7	79.3	78.0	77.5	77.0	78.3	78.1	78.5	79.5		
	2%	DB	89.9	89.6	88.8	89.2	89.7	89.7	89.8	90.7	91.3	91.4	91.5	91.5		
		MCWB	77.3	77.7	78.6	78.7	78.0	77.2	76.8	76.8	76.9	77.1	77.8	78.1		
	5%	DB	88.6	87.9	87.6	87.7	88.1	88.2	88.2	89.5	89.8	89.9	90.0	89.9		
		MCWB	76.9	77.7	78.2	78.6	78.2	77.1	76.7	76.6	76.3	76.4	76.7	76.8		
	10%	DB	87.5	86.3	86.0	86.0	87.3	87.5	87.5	87.9	88.5	89.3	89.3	89.3		
		MCWB	76.8	77.5	77.9	78.2	78.2	77.1	76.7	76.6	76.2	76.2	76.4	76.5		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	81.7	82.0	81.8	82.4	82.5	81.8	81.6	81.0	80.1	80.4	80.9	81.6		
		MCDB	86.4	86.4	85.9	86.6	87.3	86.9	86.5	87.3	88.1	88.0	88.6	88.3		
	2%	WB	80.4	80.7	80.4	80.8	80.9	80.3	80.2	79.9	79.3	79.4	79.9	80.3		
		MCDB	85.4	85.3	85.5	85.6	86.3	86.0	85.7	86.4	87.0	86.8	87.3	86.2		
	5%	WB	79.5	79.7	79.4	79.5	79.8	79.2	79.2	79.0	78.7	78.8	79.1	79.6		
		MCDB	84.7	84.3	84.3	84.3	85.1	84.8	84.8	85.6	86.2	86.2	86.2	85.9		
	10%	WB	78.8	78.9	78.8	78.8	79.0	78.5	78.5	78.3	78.0	78.2	78.5	78.9		
		MCDB	84.1	83.5	83.6	83.6	84.1	84.0	84.1	84.5	85.0	85.2	85.3	85.5		
(35) Mean Daily Temperature Range	5% DB	MDBR	10.9	10.8	10.7	10.8	11.6	12.1	12.1	12.4	12.1	11.8	11.5	11.2		
		MCDBR	11.9	12.0	12.0	12.4	12.5	12.7	12.9	13.3	12.9	12.3	12.1	11.9		
	5% WB	MCWBR	3.9	4.4	4.8	5.0	4.7	4.5	4.4	4.3	3.9	3.6	3.5	3.4		
		MCWBR	11.2	11.0	11.2	11.3	12.0	12.4	12.5	12.9	12.8	12.3	12.2	11.6		
(40) Clear-Sky Solar Irradiance	taub	taub	0.427	0.417	0.417	0.413	0.414	0.403	0.411	0.446	0.510	0.496	0.489	0.466		
		taud	2.448	2.491	2.498	2.528	2.505	2.533	2.493	2.363	2.134	2.186	2.212	2.303		
	Ebn at Noon	Ebn at Noon	288	291	288	282	274	273	272	269	259	266	268	275		
		Edh at Noon	38	36	36	34	33	32	33	39	51	49	48	43		
(44) All-Sky Solar Radiation	RadAvg	RadAvg	1569	1514	1487	1470	1528	1614	1678	1852	1920	1869	1802	1689		
	RadStd	RadStd	109	132	102	90	107	62	56	38	58	55	57	82		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	-3.31	-0.90	-1.19	N/A	N/A	N/A	N/A	N/A
(47) Regional (0 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page